JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON THURSDAY 02nd APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

PHYSICS

TEST PAPER WITH SOLUTION

SECTION-A

26. Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T) are____.

(1) $[hTLM^{-2}]$ (2) $[hT^{-1}LM^{-2}]$

(3) $[hTL^{2}M^{-2}]$ (4) $[h^{-1}T^{-1}LM^{-2}]$

Sol. $\text{Energy} = \frac{hc}{\lambda} = \frac{GM_1M_2}{r}$

$[ML^2T^{-2}] = [h]\,[T^{-1}] = [G]\,[M^2L^{-1}]$

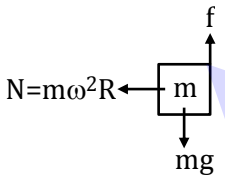
$[G] = [M^{-2}L^{+1}T^{-1}h^{+1}]$

27. A 0.5 kg mass is in contact against the inner wall of a cylindrical drum of radius 4m rotating about its vertical axis. The minimum rotational speed of the drum to enable the mass to remain stuck to the wall (without falling) is 5 rad/s. The coefficient of friction between the drum's inner wall surface and mass is_____.(Take g = 10 m/s^2)

(1) 0.1 (2) 0.5

(3) 0.7 (4) 0.3

Sol.



$N = m\omega^2R$

$f_s = mg$

$f_s \le f_\ell$

$mg \le \mu N$

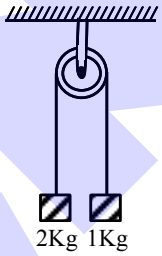
$mg \le \mu m\omega^2R$

$\mu \ge \frac{g}{\omega^2R}$

$\mu_{min} = \frac{g}{\omega^2R} = \frac{10}{10^2 \times 1} = 0.1$

28. Two blocks of masses 2 kg and 1 kg respectively, are tied to the ends of a string which passes over a light frictionless pulley as shown in the figure below. The masses are held at rest at the same horizontal level and then released. The distance traversed by the centre of mass in 2 s is____m.

(Take g = 10 m/s^2)



(1) 3.33

(2) 3.12

(3) 2.22

(4) 1.42

Sol. $a = \frac{(m_2 - m_1)}{m_1 + m_2}g = \frac{(2-1)}{3} \times 10 = \frac{10}{3}\,m/s^2$

$a_{cm} = \frac{(1)(10/3) - (2)(10/3)}{3} = \frac{-10}{9}\,m/s^2(\downarrow)$

$S_{cm} = u_{cm}t + \frac{1}{2}a_{cm}t^2$

$= 0 + \frac{1}{2}\left(\frac{10}{9}\right)(4) = \frac{20}{9}\,m\ (\text{downward})$

$S_{COM} = \frac{20}{9}\ (\text{Downward}) = 2.22$

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29. A particle having charge 10^{-9} C moving in x–y plane in fields of $0.4\ \hat{j}\ N/C$ and $4 \times 10^{-3}\ k$ T experiences a force of $\left(4\hat{i}+2\hat{j}\right) \times 10^{-10}$ N. The velocity of the particle at that instant is_____m/s.

(1) $50\hat{i}+100\hat{j}$ (2) $100\hat{i}+50\hat{j}$

(3) $-50\hat{i}+100\hat{j}$ (4) $50\hat{i}-100\hat{j}$

Sol. $\vec{F}=q\vec{E}+q\left(\vec{V}\times\vec{B}\right)$

$(4\hat{i}+2\hat{j})\times10^{-10}$

$= 10^{-9}[0.4\,\hat{j} + (v_x\,\hat{i} + v_y\,\hat{j}) \times (4 \times 10^{-3})\,\hat{k}\,]$

$(4\,\hat{i} + 2\,\hat{j}) \times 10^{-10}$

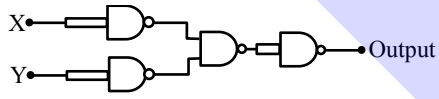
$= 10^{-9}\,(0.4\,\hat{j} - 4 \times 10^{-3}\,v_x\,\hat{j} + 4 \times 10^{-3}\,v_y\,\hat{i}\,)$

On comparing

$v_y = 100,\ v_x = 50$

$\vec{V}=50\hat{i}+100\hat{j}$

30. If X and Y are the inputs, the given circuit works as_____.



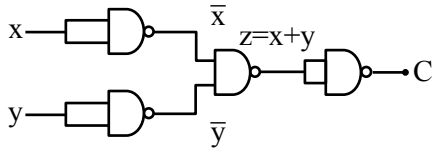
(1) OR gate

(2) AND gate

(3) NAND gate

(4) NOR gate

Sol.



$C = \overline{x+y}$

31. If a body of mass 1 kg falls on the earth from infinity, it attains velocity (v) and kinetic energy (k) on reaching the surface of earth. The values of v and k respectively are_____.

(Take radius of earth to be 6400 km and $g = 9.8 m/s^2$)

(1) 11.2 km/s; 6.27×10^7 J

(2) 11.2 km/s; 12.54×10^7 J

(3) 8.8 km/s; 6.27×10^7 J

(4) 8.8 km/s; 12.54×10^7 J

Sol. M.E.C

$K_i + U_i = K_f + U_f$

$\Rightarrow O - \frac{GM_e m}{\infty} = \frac{1}{2}mv^2 - \frac{GMm}{Re}$

$V = \sqrt{\frac{2GM}{Re}} = \sqrt{2gRe} = \sqrt{2\times9.8\times6400\times1000}$

$= 11.2$ km/s

Kinetic energy $= \frac{1}{2}mv^2$

$= 6.27 \times 10^7$ J

32. In a screw gauge the zero of main scale reference line coincides with the fifth division of the circular scale when two studs are in contact. There are 100 divisions in circular scale and pitch of screw gauge is 0.1 mm. When diameter of a sphere is measured, the reading of main scale is 5 mm and 50th division of circular scale coincides with the reference line of main scale. The diameter of sphere is _____mm.

(1) 5.045 (2) 5.055

(3) 5.450 (4) 5.550

Sol. Least count : $\frac{0.1}{100}\text{mm} = 0.001\text{mm}$

Reading = MSD + (CSD) × LC – zero error

= 5mm + 50 × 0.001 – 5 × 0.001 = 5.045 mm

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33. The surface tension of a soap bubble is 0.03 N/m. The work done in increasing the diameter of bubble from 2 cm to 6 cm is $\alpha\pi \times 10^{-4}$ J. The value of α is_____. (Take $\pi = 3.14$)

(1) 0.86 (2) 0.64

(3) 1.92 (4) 7.68

Sol. $W = \Delta U = 2T\,[4\pi R_2^2 - 4\pi R_1^2]$

$W = 2 \times 3 \times 10^{-2} \times 4\pi\,[9 \times 10^{-4} - 1 \times 10^{-4}]$

$W = 1.92\,\pi \times 10^{-4}$ J

$\alpha = 1.92$

34. A mixture of carbon dioxide and oxygen has volume 8310 cm^3, temperature 300K, pressure 100 kPa and mass 13.2 g. The number of moles of carbon dioxide and oxygen gases in the mixture respectively are_____.

(Assume both carbondioxide and oxygen gases behave like ideal gases) [R = 8.31J/mol.K]

(1) 0.15 and 0.18 (2) 0.25 and 0.08

(3) 0.21 and 0.12 (4) 0.13 and 0.20

Sol. First, we determine the total number of moles (n_{total}) in the mixture using the Ideal Gas Law: PV = nRT. Given the values:

- Pressure (P): 100 kPa = 10^5 Pa
- Volume (V): 8310 cm3 = 8.31×10^{-3} m^3
- Temperature (T): 300 K
- Gas Constant (R): 8.31 J/mol·K

$$\frac{PV}{RT} = \frac{10^5 \times 8.31 \times 10^{-3}}{8.31 \times 300} = \frac{100}{300} \approx 0.333 \text{ mole}$$

1. Mole equation: $n_1 + n_2 = 0.333$
2. Mass equation: $44n_1 + 32n_2 = 13.2$ (where 44 and 32 are the molar masses of CO_2 and O_2 respectively)
3. Solve for individual moles

   Using the first equation to express n_2 as $0.333 - n_1$, we substitute this into the mass equation:

   $44n_1 + 32(0.333 - n_1) = 13.2$

   $44n_1 + 10.667 - 32n_1 = 13.2$

   $12n_1 = 2.533$

   $n_1 \approx 0.211$ mol

   Then, calculate n_2 :

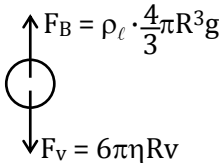
   $n_2 = 0.333 - 0.211 = 0.122$ mol

35. If an air bubble of diameter 2 mm rises steadily through a liquid of density 2000 kg/m^3 at a rate of 0.5 cm/s, then the coefficient of viscosity of liquid is_____Poise. (Take g = 10 m/s^2)

(1) 0.88 (2) 8.8

(3) 88.8 (4) 0.088

Sol.



For constant speed

$F_B = F_v$

$$\Rightarrow \rho_\ell \frac{4}{3}\pi R^3 g = 6\pi\eta Rv$$

$$\frac{2}{9}\frac{\rho_\ell R^2 g}{v} = \eta$$

$$\Rightarrow \frac{2}{9} \times \frac{2000 \times 10^{-6} \times 10}{5 \times 10^{-3}} = \eta$$

$$\Rightarrow \eta = \frac{8}{9} \times 10$$

$$\Rightarrow \eta = \frac{80}{9} \text{ Poise} = 8.88$$

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36. A spherical ball of mass 2 kg falls from a height of 10 m and is brought to rest after penetrating 10 cm into sand. The average force exerted by sand on the ball is _____N.

(1) 1980 (2) 2020

(3) 2000 (4) 1000

Sol. $(W.D.)_g + (W.D.)_{sand} = \Delta K$

$$2\times10\times\left(10+\frac{10}{100}\right)-F_{sand}\times\left(\frac{10}{100}\right)=0 \ =0$$

$$F_{sand} = 2(10 + 0.1)\times100 = 2020$$

37. An electromagnetic wave travels in free space along the *x*-direction. At a particular point in space and time, $\vec{B}=2\times10^{-7}\hat{j}$ T is associated with this wave. The value of corresponding electric field $\vec{E}$ at this point is _____V/m.

(1) $60\,\hat{k}$ (2) $-60\,\hat{k}$

(3) $30\,\hat{k}$ (4) $-600\,\hat{k}$

Sol. $\vec{E}=\vec{B}\times\vec{C}$

$$=2\times10^{-7}\hat{j}\times3\times10^{8}\hat{i}$$

$$\vec{E}=-60\hat{k}\ V/m$$

38. Two resistors of 200 Ω and 400 Ω are connected in series with a battery of 100 V. A bulb rated at 200 V, 100 W is connected across the 400 Ω resistance. The potential drop across the bulb is_____V.

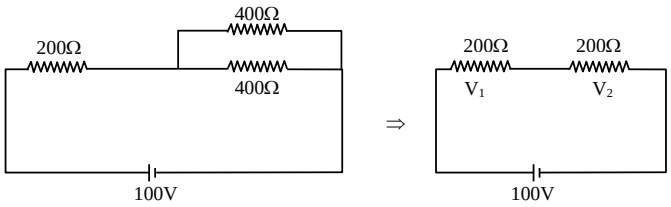
(1) 25 (2) 50

(3) 66.6 (4) 100

Sol. For resistance of bulb

$$R=\frac{V^2}{P}=\frac{200\times200}{100}=400\Omega$$

Now circuit will look like this



V_1 & V_2 will be same as resistances are same

$V_1 = V_2 = 50V$

$\boxed{V_{bulb} = V_2 = 50V}$

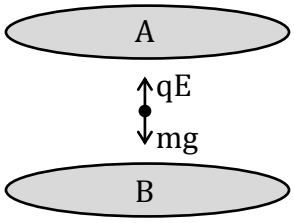
39. Two metal plates (A,B) are kept horizontally with separation of $\left(\frac{12}{\pi}\right)$ cm, with plate A on the top. An atomizer jet sprays oil (density 1.5 g/cm^3) droplets of radius 1mm horizontally. All oil droplets carry a charge 5 nC. The potentials V_A and V_B are required on plates A and B respectively in order to ensure the droplets do not descend. The values of V_A and V_B are_____.

(Neglect the air resistance to the droplets and take g = 10 m/s^2)

(1) 100 V and 580 V (2) 580 V and 100 V

(3) 60 V and 400 V (4) 0 V and –200 V

Sol.



$$\frac{\Delta V}{(12/\pi)\times10^{-2}}\times5\times10^{-9}=\frac{4}{3}\pi(r^3)\rho(g)$$

$$\Delta V\times5\times10^{-7}=16(r^3)\rho(g)$$

$\Delta V = 480$, $V_A < V_B$

Plate A will be lower potential.

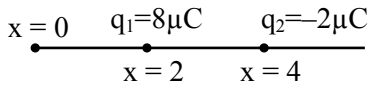
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40. Two point charges 8μC and –2μC are located at x = 2 cm and x = 4 cm, respectively on the x-axis. The ratio of electric flux due to these charges through two spheres of radii 3 cm and 5 cm with their centers at the origin is_____.

(1) 4 : 1 (2) 3 : 4

(3) 4 : 3 (4) 4 : 5

Sol.



For sphere with radius 3cm and center origin

$$\phi_1 = \frac{q_{en}}{\varepsilon_0} = \frac{8\mu C}{\varepsilon_0}$$

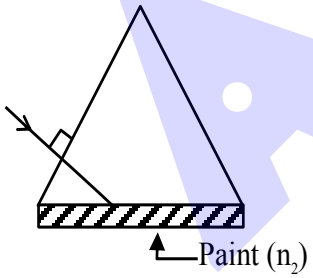
for sphere of radius 5cm

$$\phi_2 = \frac{q_{en}}{\varepsilon_0} = \frac{(8-2)\mu C}{\varepsilon_0} = \frac{6\mu C}{\varepsilon_0}$$

$$\frac{\phi_1}{\phi_2} = \frac{8}{6} = \frac{4}{3} = 1.33$$

$$\frac{\phi_1}{\phi_2} = 1.33$$

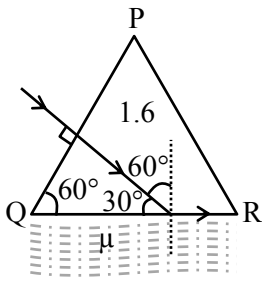
41. One side of an equilateral prism is painted by a transparent material of refractive index n_2. The refractive index of prism is 1.6. The minimum value of n_2 required for total internal reflection from painted face is_____.



(1) $\frac{3\sqrt{3}}{1.6}$ (2) $\sqrt{3}$

(3) $\frac{3.2}{\sqrt{3}}$ (4) $\frac{4\sqrt{3}}{5}$

Sol.



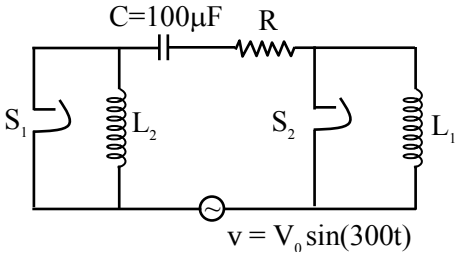
For surface QR

i = 60°

1.6 sin 60° = μ sin 90°

$$\mu = 1.6 \times \frac{\sqrt{3}}{2} = \frac{4}{5} \times \sqrt{3}$$

42. The figure given below shows an LCR series circuit with two switches S_1 and S_2. When switch S_1 is closed keeping S_2 open, the phase difference (ϕ) between the current and source voltage is 30° and phase difference is 60° when S_2 is closed keeping S_1 open. The value of $(3L_1 - L_2)$ is___ H.



(1) $\frac{9}{2}$

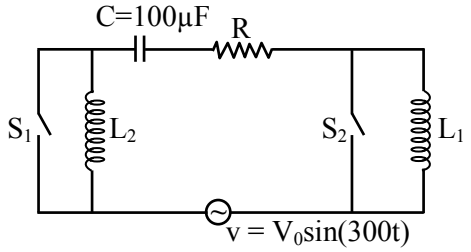
(2) $\frac{2}{9}$

(3) $\frac{1}{3}$

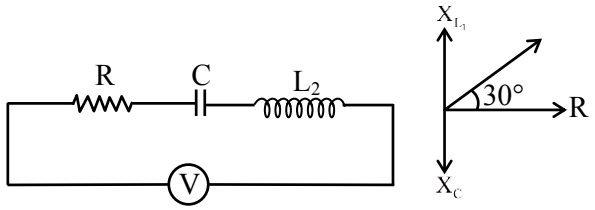
(4) 3

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Sol.



For S_1 is closed and S_2 open

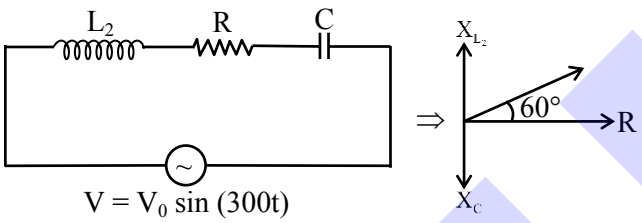


$V = V_0 \sin(300\,t)$

$$\frac{X_{L_1} - X_C}{R} = \tan(30°) = \frac{1}{\sqrt{3}}$$

$$\sqrt{3}\,(X_{L_1} - X_C) = R \quad .....(1)$$

For S_1 is open and S_2 is closed.



$$\frac{|X_{L_2} - X_C|}{R} = \tan 60° = \sqrt{3}$$

$$\frac{|X_{L_2} - X_C|}{\sqrt{3}} = R \quad ....(2)$$

from Eq. (1) and (2)

$$\sqrt{3}\,(X_{L_1} - X_C) = \frac{(X_{L_2} - X_C)}{\sqrt{3}}$$

$$3\left(\omega L_1 - \frac{1}{\omega C}\right) = \left(\omega L_2 - \frac{1}{\omega C}\right)$$

$$|3L_1 - L_2| = \frac{3}{\omega^2 C} - \frac{1}{\omega^2 C} = \frac{2}{9\times10^4 \times 100 \times 10^{-6}}$$

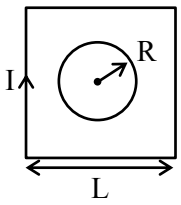
$$|3L_1 - L_2] = \frac{2}{9}$$

43. A circular current loop of radius R is placed inside square loop of side length L(L>>R) such that they are co-planar and their centers coincide. The permeability of free space is μ_0. The mutual inductance between circular loop and square loop is____.

(1) $2\sqrt{2}\frac{\mu_0 L^2}{R}$ (2) $\sqrt{2}\frac{\mu_0 L^2}{R}$

(3) $\sqrt{2}\frac{\mu_0 R^2}{L}$ (4) $2\sqrt{2}\frac{\mu_0 R^2}{L}$

Sol.



Magnetic field due to square

$$B = 4 \times \frac{\mu_0 I}{4\pi\frac{L}{2}}\ \{2\sin 45°\}$$

$$= \frac{4\mu_0 I}{\pi L} \times \frac{1}{\sqrt{2}} = \frac{2\sqrt{2}\mu_0 I}{\pi L}$$

$$Q = B \times A_{circle}$$

$$Q = \frac{2\sqrt{2}\mu_0 I}{\pi L} \times \pi R^2$$

$$Q = \frac{2\sqrt{2}\mu_0 R^2}{L} I$$

$$\text{Inductance} = \frac{2\sqrt{2}\mu_0 R^2}{L}$$

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44. The binding energy per nucleon of $^{209}_{83}Bi$ is _____ MeV.

[Take $m\left(^{209}_{83}Bi\right)$ = 208.980388 u, m_p = 1.007825 u, m_n = 1.008665 u, 1u = 931 MeV/c^2]

(1) 7.48 (2) 7.84

(3) 8.79 (4) 6.94

Sol. $83\,{}_1P^1 + 126\,{}_0n^1 \rightarrow {}_{83}Bi^{209}$

$\Delta m = 83m_p + 126\,m_n - M_{Bi}$

$= 83 \times 1.007825 + 126 \times 1.008665 - 208.980388$

$= 1.760877$ u

BE = 1.760877 × 931 MeV = 1639.3764

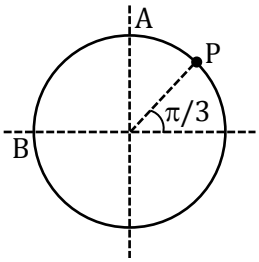
$\frac{BE}{A} = \frac{1639.3764}{209}$ MeV = 7.8439 MeV

45. The equation of motion of a particle is given by $x = a\sin(50t + \pi/3)$cm. The particle will come to rest at time t_1 and it will have zero acceleration at time t_2. The t_1 and t_2 respectively are_____.

(1) $\frac{\pi}{300}$s, $\frac{\pi}{75}$s (2) $\frac{\pi}{75}$s, $\frac{\pi}{300}$s

(3) $\frac{\pi}{300}$s, $\frac{\pi}{25}$s (4) $\frac{\pi}{50}$s, $\frac{\pi}{100}$s

Sol. Phase of the motion of particle is as follows :



V = 0, when particle cross A.

$\therefore t_1 = \frac{\pi}{6\omega} = \frac{\pi}{300}$

a = 0, when particle cross B.

$\therefore t_2 = \frac{2\pi}{3\omega} = \frac{\pi}{75}$

SECTION-B

46. In a Young's double slit experiment, the intensity at some point on the screen is found to be $\frac{3}{4}$ times of the maximum of the interference pattern. The path difference between the interfering waves at this point is $\frac{\lambda}{x}$ where λ is wavelength of the incident light. The value of x is _____.

Sol. $I_r = I_{max}\cos^2\frac{\phi}{2}$

$\frac{3I_{max}}{4} = I_{max}\cos^2\frac{\phi}{2}$

$\cos\frac{\phi}{2} = \frac{\sqrt{3}}{2}$

$\frac{\phi}{2} = 30° = \frac{\pi}{6}$

$\left(\phi = \frac{\pi}{3}\right)$

Path difference $\Delta x = \frac{\phi}{2\pi} \times \lambda$

$\Rightarrow \frac{\pi}{3 \times 2\pi} \times \lambda$

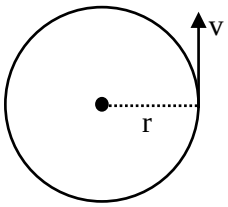
$\Delta x = \frac{\lambda}{6}$

x = 6

47. Using Bohr's model, calculate the ratio of the magnetic fields generated due to the motion of the electrons in the 2nd and 4th orbits of hydrogen atom__.

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Sol.



$$B = \frac{\mu_0 I}{2\pi r}$$

$$I = \frac{ev}{2\pi r}$$

$$B \propto \frac{V}{r^2}$$

$$V \propto \frac{Z}{n}$$

$$r \propto \frac{n^n}{Z}$$

$$B \propto \frac{\left(\frac{Z}{n}\right)}{\left(\frac{n^2}{Z}\right)^2}$$

$$B \propto \frac{Z^3}{n^5}$$

$$\frac{B_1}{B_2} = \frac{n_2^5}{n_1^5} = \frac{4^5}{2^5} = 64$$

48. 5 moles of unknown gas is heated at constant volume from 10°C to 20°C. The molar specific heat of this gas at constant pressure c_p = 8 cal/mol. °C and R = 8.36 J/mol.°C. The change in the internal energy of the gas is ____ calorie.

Sol. $C_P - C_V = R \Rightarrow C_V = (C_P - R)$

$C_V = 6$ cal/mol°C

$\Delta U = nC_V \Delta T$

$= 5 \times 6 \times 10 = 300$ cal

49. If sunlight is focused on a paper using convex lens, it starts burning the paper in shortest time when the lens is kept at 30 cm above the paper. If the radius of curvature of the lens is 60 cm then the refractive index of the lens material is $\frac{\alpha}{10}$. The value of α is___ .

Sol. To burn in minimum time distance of paper must be at focus.

So f = 30 cm

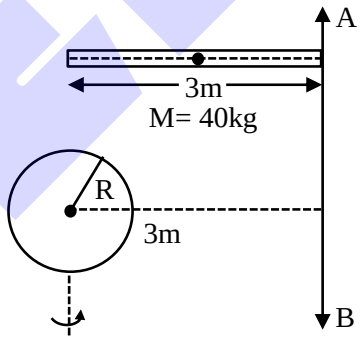
$$\frac{1}{30} = (\mu - 1)\left(\frac{1}{60} - \frac{1}{-60}\right)$$

$$\frac{1}{30} = \frac{\mu - 1}{30}$$

$\mu = 2$

$\boxed{\alpha = 20}$

50. Moment of inertia about an axis AB for a rod of mass 40 kg and length 3 m is same as that of a solid sphere of mass of 10 kg and radius R about an axis parallel to AB axis with separation of 3 m as shown in figure below. The value of R is given as $\sqrt{\frac{\alpha}{2}}$. The value of α is_____.



Sol. $I_{Rod} = I_{SS}$

$$\frac{M_1 L^2}{3} = \left(\frac{2}{5} m_2 R^2 + m_2 L^2\right)$$

$$\frac{40 \times (3)^2}{3} = \frac{2}{5} \times 10 \times R^2 + 10 \times (3)^2$$

$120 = 4R^2 + 90$

$4R^2 = 30$

$$R = \sqrt{\frac{15}{2}} = \sqrt{\frac{\alpha}{2}}$$

$\alpha = 15$

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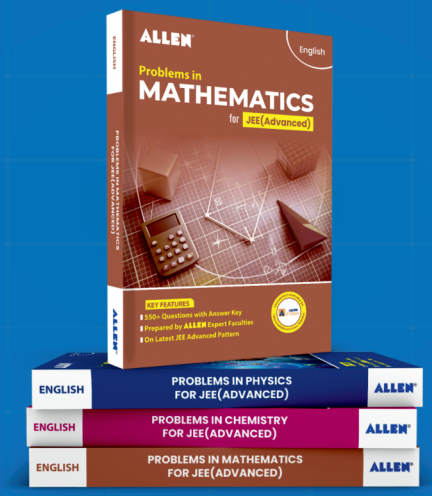


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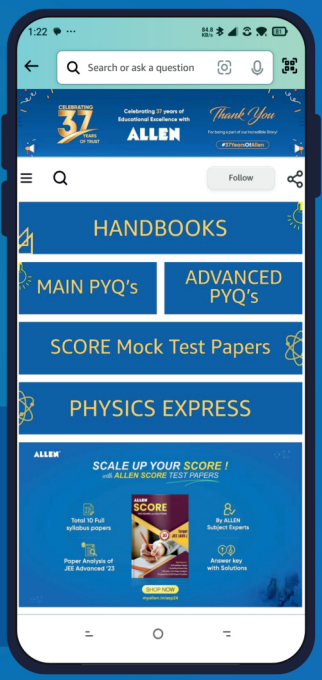
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